

# Financial Management: An Overview

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## LEARNING OBJECTIVES

After studying this chapter, you should be able to:

- List the pros and cons of different forms of business organisation
- Explain why a firm should seek to maximise shareholder wealth
- Cite the key responsibilities of financial managers
- Explain the fundamental principle of finance
- Show why agency costs may arise in companies and what can be done to mitigate them
- Explain the relationship of accounting and economics to finance

Suppose you are planning to start your own business. No matter what the nature of your proposed business is and how it is organised, you will have to address the following questions:

- What capital investments should you make? That is, what kinds of real estate, machineries, R & D programmes, IT infrastructure, and so on should you invest in?
- How will you raise money to pay for the proposed capital investments? That is, what will be the mix of equity and debt in your financing plan?
- How will you handle the day-to-day financial activities like managing your inventories, collecting your receivables, and paying your suppliers?

While these are not the only concerns of financial management, they are certainly the central ones.

This book discusses the theories, analytical methods, and practical considerations that are helpful in addressing various issues in financial management, a discipline that has assumed great significance in recent times.

Before we begin our long journey, let us get an overview of financial management, also referred to as *corporate finance* or *managerial finance*. This chapter provides the overview.

## 1.1 FORMS OF BUSINESS ORGANISATION

All firms face the basic problems of capital budgeting, capital structure, dividend policy, working capital management, and financial control. However, these issues tend to be more complex for companies than for other forms of organisation.

Since this book focuses primarily on financial management of companies—note that large firms are almost invariably organised as companies—you should know how a company differs from other forms of business organisation like sole proprietorship, partnership, and cooperative society.

### Sole Proprietorship

A sole proprietorship firm is a business owned by a single person. This is the simplest form of business, subject to minimal regulation. You can set up a sole proprietorship firm by obtaining a license, if the same is required for the business you want to engage in, and throwing open your doors. Thanks to its simplicity, most businesses begin as sole proprietorship firms. No wonder there are more sole proprietorships than any other form of organisation.

From a legal and tax point of view, a sole proprietorship firm has no separate status apart from its owner. The owner realises all the profits and bears all the losses. The owner indeed has unlimited personal liability for the debts of the business. By the same token, there is no distinction between business and personal income and all business income is taxed as personal income.

The equity capital of a sole proprietorship is limited to the personal wealth of the owner. Hence such firms often cannot grow beyond a point for want of capital. It may be somewhat difficult to transfer the ownership of a sole proprietorship firm as it involves sale of the entire business to the buyer.

### Partnership

A partnership firm is a business owned by two or more persons. It may be viewed as an extension of sole proprietorship. The partners bear the risks and reap the rewards of the business.

Generally, a partnership comes into being with the execution of a partnership deed that specifies, *inter alia*, the capital contributions, shares, rights, duties, and obligations of the partners. In India, partnerships are governed by the Indian Partnership Act, 1932. This legislation regulates the relationship between the partners *inter se* as well as between the partners and the parties dealing with the partnership firm.

A partnership firm is a distinct legal and tax entity. It can pay interest and remuneration to the partners and claim the same as tax-deductible expenses. Of course, these incomes are taxable in the hands of the partners. The tax rate applicable to the net profit of the partnership firm is 30 percent.

While a partnership firm can benefit from the varied experience and expertise of the partners and draw on their combined capital resources, its advantages and disadvantages are more or less similar to that of a sole proprietorship firm.

**Limited Liability Partnership** Recently a new form of business organisation called Limited Liability Partnership (LLP) was introduced in India. Its distinctive feature is that it is a partnership firm wherein the liability of the partners is limited. An LLP must have a minimum of two partners and at least one of them should be an Indian resident. The partners are accountable for regulatory and legal compliance. The rights and duties of the partners are governed by the agreement between the partners or between the LLP and the partners.

Since the LLP is treated as a firm, it does not have to pay the minimum alternative tax of 18.5 percent of book profits and the dividend distribution tax. The interest that an LLP can pay on the investments made by the partners is limited to 12 percent of the total income of the LLP.

The remuneration can be paid to the partners as per the slabs fixed under the law. The net profit of the LLP would be taxed at 30 percent. The partners, of course, have to pay taxes for their interest and remuneration received from the LLP.

## **Cooperative Society**

A cooperative society may be defined as 'a society which has as its objective the promotion of economic interests of its members in accordance with cooperative principles.'

The key features of a cooperative organisation are as follows: (a) While there is no maximum limit for membership, a minimum of ten members are required to form a cooperative society. The members of a cooperative society are its owners. (b) The management of a cooperative society is vested in the hands of the managing committee elected by members on the principle of 'one member, one vote'. (c) The dividend payable on the capital contributed by members is subject to a ceiling of 9 percent. The surplus left after the dividend payment is distributed in the form of bonus which is linked to the volume of business done by members with the society.

The advantages of a cooperative organisation are as follows: (a) It can be formed easily. (b) The liability of the members is limited. (c) Grants and financial assistance are provided by the government to cooperative organisations.

The disadvantages of a cooperative organisation are as follows: (a) Cooperatives cannot ordinarily employ outside talent. (b) Members do not have an incentive to provide capital because the dividend rate is low and the principle of 'one member, one vote' is followed. (c) Often, influential members exploit the cooperative society for personal gains.

## **Company**

A company is collectively owned by the shareholders who entrust the task of management to their elected representatives called the directors. The salient features of a company are as follows:

- The company is a distinct legal 'person,' separate from its owners, the shareholders. It can own assets, incur liabilities, enter into contracts, sue and be sued in its name.

## 1.6 *Fundamentals of Financial Management*

- The liability of the shareholders of a company is limited to the share capital subscribed to by them. Once this amount is fully paid up, they have no further obligation.
- A company must pay taxes on its profits. Moreover, shareholders of the company are liable to pay taxes on the dividend received by them.<sup>1</sup> So, in effect, there is double taxation.
- Setting up and managing a company is more complicated than setting up and managing other forms of organisation because companies are governed by the Indian Companies Act, a very elaborate and comprehensive piece of legislation.

A company may be a private limited company or a public limited company. The key differences between them are as follows:

- A private limited company must have at least two shareholders (members) whereas a public limited company must have at least seven shareholders. While there is no limit on the number of shareholders of a public limited company, the number of shareholders of a private company cannot exceed two hundred.
- A public limited company invites members of the public to subscribe to its shares, whereas a private limited company cannot do so.
- A public limited company permits free transfer of shares whereas a private limited company usually imposes restrictions on such transfers.

On the whole, the public limited company is the most appropriate form of business organisation, except, of course, when the business is small. The reasons are: (a) The risk to investors is limited. (b) The potential for growth is immense because of access to substantial funds. (c) Investors enjoy liquidity because of free transferability. Thanks to these advantages, large and medium-sized businesses are generally organised as public limited companies. Reliance Industries, State Bank of India, Ranbaxy Laboratories, and Infosys Technologies, for example, are public limited companies. So are overseas businesses such as General Electric, Intel, British Petroleum, Sony, and Asea Brown Boveri.

## 1.2 FINANCIAL DECISIONS IN A FIRM

As mentioned in the beginning of this chapter, there are three broad areas of financial decision making viz., capital budgeting, capital structure, and working capital management.

### Capital Budgeting

The first and perhaps the most important decision that any firm has to make is to define the business or businesses that it wants to be in. This is referred to as strategic planning and it has a significant bearing on how capital is allocated in the firm. As strategic

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<sup>1</sup> Presently in India, however, shareholders are not liable to pay any tax on the dividend received by them. Rather, the company itself has to pay taxes on the dividend paid by it. This is, of course, in addition to the taxes it pays on its profits.

planning calls for evaluating costs and benefits spread out over time, it is essentially a financial decision making process.

Once the managers of a firm choose the business or businesses they want to be in, they have to develop a plan to invest in buildings, machineries, equipments, research and development, godowns, showrooms, distribution network, information infrastructure, brands, and other long-lived assets. This is the capital budgeting process.

Considerable managerial time, attention, and energy are devoted to identify, evaluate, and implement investment projects. When you look at an investment project from the financial point of view, you should focus on the magnitude, timing, and riskiness of cash flows associated with it. In addition, consider the options embedded in the investment projects.

## **Capital Structure**

Once a firm has decided on the investment projects it wants to undertake, it has to figure out ways and means of financing them.

The key issues in capital structure decision are: What is the optimal debt-equity ratio for the firm? Which specific instruments of equity and debt finance should the firm employ? Which capital markets should the firm access? When should the firm raise finances? At what price should the firm offer its securities?

An allied issue is the distribution policy of the firm. What is the optimal dividend payout ratio for the firm? Should the firm buyback its own shares?

Capital structure and dividend decisions should be guided by considerations of cost and flexibility, in the main. The objective should be to minimise the cost of financing without impairing the ability of the firm to raise finances required for value creating investment projects.

## **Working Capital Management**

Working capital management, also referred to as short-term financial management, refers to the day-to-day financial activities that deal with current assets (inventories, debtors, short-term holdings of marketable securities, and cash) and current liabilities (short-term debt, trade creditors, accruals, and provisions).

The key issues in working capital management are: What is the optimal level of inventory for the operations of the firm? Should the firm grant credit to its customers and, if so, on what terms? How much cash should the firm carry on hand? Where should the firm invest its temporary cash surpluses? What sources of short-term finance are appropriate for the firm?

## **1.3 GOAL OF THE FIRM**

In proprietorships and partnerships owners are actively involved in management. But in companies, particularly large public limited companies, which have many shareholders, ownership is separated from management. For example, it is practically impossible for tens of thousands shareholders of Larsen and Toubro to participate actively in management. They have to necessarily delegate authority to the board of directors which in turn appoints the top management.

## 1.8 Fundamentals of Financial Management

Since shareholders differ in their tastes, wealth, investment horizons, and personal opportunities, delegation can work only if they can agree on a common objective. Indeed, there is a natural financial objective on which almost all shareholders would agree: maximise the current market value of the firm's equity. Every shareholder, regardless of his personal tastes or preferences, can do more if his shares are worth more. He or she can give money for charity or travel to exotic locations; he or she can spend now or save for future.

The goal of shareholder wealth maximisation has been eloquently defended by distinguished finance scholars, economists, and practitioners.

"Should a firm maximise the welfare of employees, or customers, or creditors? These are bogus questions. The real question is: What should a firm do to maximise its contribution to the society? The contribution to the society is maximised by maximising the value of the firm."<sup>2</sup>

"The quest for value drives scarce resources to their most productive uses and their most efficient users. The more effectively resources are deployed, the more robust will be the economic growth and the rate of improvement in our standard of living. Adam Smith's 'invisible hand' is at work when investors' private gain is a public value."<sup>3</sup>

You may ask why are we overlooking the interests of customers, employees, and suppliers. No company can succeed unless it offers value for money to its customers, provides satisfactory employment conditions to its employees, and treats its suppliers fairly. Hence the question of neglecting the interests of these constituencies to promote the welfare of shareholders simply does not arise. More value for shareholders does not imply less value for customers, employees, or suppliers. On the contrary, only by focusing on creating value for its shareholders can a firm ensure that it has durable and mutually beneficial relationships with its customers, employees, and suppliers.

This view has been endorsed by eminent practitioners. As Jack Welch, arguably the most outstanding CEO of our times, said: 'A proper balance between shareholders, employees, and communities is what we try to achieve. But it is a tough balancing act because, in the end, if you don't satisfy shareholders, you don't have the flexibility to do the things you have to do to take care of employees or communities. In our society, whether we like it or not, we have to satisfy shareholders.'

### Alternative Goals

Are there other goals, besides the goal of maximal shareholder wealth, expressing the shareholders' viewpoint? Several alternatives have been suggested: maximisation of profit, maximisation of earnings per share, maximisation of return on equity (defined as equity earnings/net worth). Let us examine them.

Maximisation of profit is not as inclusive a goal as maximisation of shareholder wealth. It suffers from several limitations:

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<sup>2</sup> Michael Jensen, *Economist*, 1997.

<sup>3</sup> Bennett Stewart, *The Quest for Value: A Guide to Senior Management*, New York: Harper & Row, 1991.

- Profit in absolute terms is not a proper guide to decision making. It should be expressed either on a per share basis or in relation to investment.
- It leaves considerations of timing and duration undefined. There is no guide for comparing profit now with profit in future or for comparing profit streams of different durations.
- It glosses over the factor of risk. It cannot, for example, discriminate between an investment project which generates a certain profit of ₹50,000 and an investment project which has a variable profit outcome with an expected value of ₹50,000.
- The calculation of profit is influenced by the accounting rules employed.

The goals of maximisation of earnings per share and maximisation of return on equity do not suffer from the first limitation mentioned above. However, they do suffer from the other limitations and hence are not suitable.

In view of the shortcomings of the alternatives discussed above, maximisation of the wealth of equity shareholders (as reflected in the market value of equity) appears to be the most appropriate goal for financial decision making.

## 1.4 FUNDAMENTAL PRINCIPLE OF FINANCE

The key question that you have to ask before making a business decision is: will the decision raise the market value of the firm? To answer this question, let us look at the fundamental principle of finance:

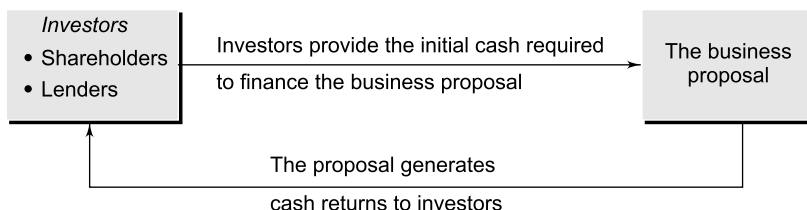
**A business proposal—regardless of whether it is a new investment or acquisition of another company or a restructuring initiative—raises the value of the firm only if the present value of the future stream of net cash benefits expected from the proposal is greater than the initial cash outlay required to implement the proposal.**

The difference between the present value of future cash benefits and the initial outlay represents the net present value or NPV of the proposal:

Net present value = Present value of future cash benefits – Initial cash outlay

Note that the costs and benefits of a business proposal have to be measured in cash. As shown in Exhibit 1.1, investors who finance a proposal invest cash and hence are interested only in cash returns.

**Exhibit 1.1** Cash Alone Matters



### 1.10 Fundamentals of Financial Management

To convert the expected cash returns from the proposal into a present value figure, an appropriate discount rate has to be applied. The discount rate reflects the riskiness of the proposal.

## 1.5 ORGANISATION OF THE FINANCE FUNCTION

Financial management is in many ways an integral part of the jobs of managers who are involved in planning, allocation of resources, and control. The responsibilities for financial management are dispersed throughout the organisation. For example:

- The engineer, who proposes a new plant, shapes the investment policy of the firm.
- The marketing analyst provides inputs in the process of forecasting and planning.
- The purchase manager influences the level of investment in inventories.
- The sales manager has a say in the determination of the receivables policy.
- Departmental managers, in general, are important links in the financial control system of the firm.

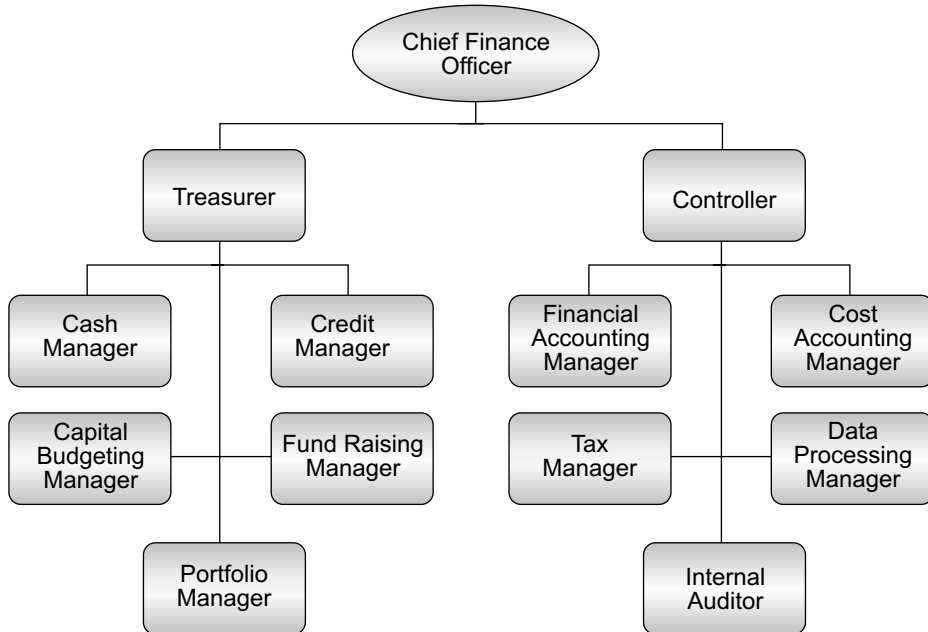
There are, however, many tasks of financial management and allied areas (like accounting) which are specialised in nature and which are attended to by specialists. These tasks and their typical distribution between the two key financial officers of the firm, the treasurer and the controller, are shown in Exhibit 1.2. Note that the treasurer is responsible mainly for financing and investment activities and the controller is concerned primarily with accounting and control.

**Exhibit 1.2** Functions of the Treasurer and the Controller

| <i>Treasurer</i>      | <i>Controller</i>     |
|-----------------------|-----------------------|
| Obtaining finance     | Financial accounting  |
| Banking relationship  | Internal auditing     |
| Cash management       | Taxation              |
| Credit administration | Management accounting |
| Capital budgeting     | Management control    |

Typically, the Chief Finance Officer supervises the work of the treasurer and the controller. In turn, these officers are assisted by several specialist managers working under them. The finance function in a large organisation may be structured as shown in Exhibit 1.3.

The finance officers, in addition to their specialised responsibilities, have significant involvement in injecting financial discipline in corporate management processes. They are responsible for emphasising upon the need for rationality in the use of funds and the need for monitoring the operations of the firm to achieve desired financial results. In this respect the tasks of financial officers have assumed new dimensions. Instead of just looking after routine financing and accounting activities, they guide and participate in the tasks of planning, funds allocation, and control so that the financial point of view is sufficiently emphasised upon in the process of corporate management.

**Exhibit 1.3** Organisation of Finance Function

## 1.6 RELATIONSHIP OF FINANCE TO ECONOMICS AND ACCOUNTING

Financial management has a close relationship to economics on the one hand and accounting on the other.

### Relationship to Economics

There are two important linkages between economics and finance. The macro-economic environment defines the setting within which a firm operates and the micro-economic theory provides the conceptual underpinning for the tools of financial decision making.

Key macro-economic factors like the growth rate of the economy, the domestic savings rate, the role of the government in economic affairs, the tax environment, the nature of external economic relationships, the availability of funds to the corporate sector, the rate of inflation, the real rate of interest, and the terms on which the firm can raise finances define the environment in which the firm operates. No finance manager can afford to ignore the key developments in the macro-economic sphere and the impact of the same on the firm.

While an understanding of the macro-economic developments sensitises the finance manager to the opportunities and threats in the environment, a firm grounding in micro-economic principles sharpens his analysis of decision alternatives. Corporate finance, in essence, is applied micro-economics. For example the principle of marginal analysis—a key principle of micro-economics according to which a decision should be guided by a

### 1.12 Fundamentals of Financial Management

comparison of incremental benefits and costs—is applicable to a number of managerial decisions in finance.

To sum up, a basic knowledge of macro-economics is necessary for understanding the environment in which the firm operates and a good grasp of micro-economic principles is helpful in sharpening the tools of financial decision making.

### Relationship to Accounting

The finance and accounting functions are closely related and almost invariably fall within the domain of the Chief Finance Officer as shown in Exhibit 1.3. Given this affinity, it is not surprising that in popular perception finance and accounting are often considered indistinguishable or at least substantially overlapping. However, as a student of finance you should know how the two differ and how the two relate. The following discussion highlights the differences and relationship between the two.

**Score Keeping vs. Value Maximising** Accounting is concerned with score keeping, whereas finance is aimed at value maximising. The primary objective of accounting is to measure the performance of the firm, assess its financial condition, and determine the base for tax payment. The principal goal of financial management is to create shareholder value by investing in positive net present value projects and minimising the cost of financing. Of course, financial decision making requires considerable inputs from accounting. As Gitman says:

“The accountant’s role is to provide consistently developed and easily interpreted data about the firm’s past, present, and future operations. The financial manager uses these data, either in raw form or after certain adjustments and analyses, as an important input to the decision-making process.”

**Accrual Method vs. Cash Flow Method** The accountant prepares the accounting reports based on the accrual method which recognises revenues when the sale occurs (irrespective of whether the cash is realised immediately or not) and matches expenses to sales (irrespective of whether cash is paid or not). The focus of the finance manager, however, is on cash flows. He is concerned about the magnitude, timing, and risk of cash flows as these are the fundamental determinants of value.

**Certainty vs. Uncertainty** Accounting deals primarily with the past. It records what has happened. Hence it is relatively more objective and certain. Finance is concerned mainly with the future. It involves decision making under imperfect information and uncertainty. Hence it is characterised by a higher degree of subjectivity.

## 1.7 REGULATORY FRAMEWORK

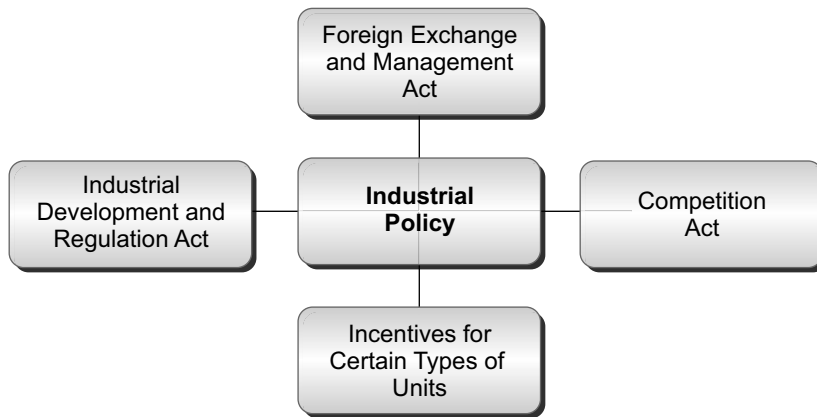
Corporate investment and financing decisions are guided, shaped, and circumscribed by a fairly comprehensive regulatory framework which seeks to (i) define avenues of investment available to business firms in different categories, ownership-wise and size-wise, (ii) induce investments along certain lines by providing incentives, concessions, and reliefs, and (iii) impose restrictions on the ways and means by which business firms can raise and deploy funds. The principal elements of this regulatory framework are:

- Industrial Policy
- Companies Act
- Securities and Exchange Board of India Guidelines

## Industrial Policy

The industrial policy of the government encompasses legislations (like the Industrial Development and Regulation Act) and incentives (like the incentives for export-oriented units) which are applicable to the industrial sector as shown in Exhibit 1.4.

**Exhibit 1.4** Industrial Policy



In the past, the industrial policy framework of the government was characterised by extensive bureaucratic control over choices which ought to have been the prerogative of the entrepreneurs. Accordingly, over time, bureaucratic controls became dysfunctional and led to dilatory decision making, corruption, and inefficient choices of scale, location, and technology. Even though some relaxation was effected over the years, the basic structure remained more or less unchanged. The New Industrial Policy of 1991, announced in the wake of liberalisation initiatives of the government, however, represented a significant departure from the earlier regime. A watershed in India's industrial policy, it sought to correct the distortions or weaknesses that had crept in, unshackle the Indian industry from the multiplicity of administrative and legal controls, and enhance the international competitiveness of Indian industry.

## Companies Act

The Companies Act 2013, together with its amendments, is a very comprehensive piece of legislation, regulating the functioning of joint stock companies. As far as corporate financial management is concerned, the Companies Act has provisions covering matters like types of share capital, issue of share capital, issue of debentures, loans, investments, inter-corporate investments, distribution of dividends, convertible loans, reorganisation, amalgamation, and liquidation.

### 1.14 *Fundamentals of Financial Management*

Some of the important company law provisions pertaining to financial management are:

1. A company can issue only two kinds of shares: equity shares and preference shares.
2. Additional issue of equity shares has to be offered to existing equity shareholders in proportion to the shares held by them unless the company, in a general body meeting, decides otherwise by passing a special resolution, or by passing an ordinary resolution and securing the permission of the central government.
3. Share capital issue cannot be made publicly unless a prospectus, giving prescribed information about the company, is furnished.
4. Debentures carrying voting rights cannot be issued.
5. The board of directors of a company or a subsidiary thereof shall not, except with the consent of the company in a general meeting, borrow money which, together with those already borrowed by the company (apart from temporary loans obtained from the company's bankers in the ordinary course of business), exceeds the aggregate of the paid-up capital of the company and its free reserves.
6. A company can buy up to 10 per cent of the subscribed capital of another corporate body, provided that the aggregate of investments made in all other corporate bodies does not exceed 30 per cent of the subscribed capital of the investing company.
7. A company can lend only up to 30 per cent of its paid-up capital and free reserves to other companies where the lending company and the borrowing company are not under the same management and only up to 20 per cent where they are under the same management.
8. Dividends are payable only out of profits after setting aside a certain percentage towards reserves.
9. A company is required to prepare its financial statements (statement of profit and loss and balance sheet) in a certain manner and format and get the same audited by a chartered accountant.
10. A public company is required to get its audited financial statements approved by its shareholders. (The financial statements along with the Directors' Report, Auditor's Report, and annexures to the financial statements as prescribed by the Companies Act constitute the annual report of the company.)

### **SEBI Guidelines**

In 1989, SEBI was created by an administrative fiat of the Ministry of Finance. Since then, SEBI has gradually been granted more and more powers. With the repeal of the Capital Issues Control Act and the enactment of the SEBI Act in 1992, the regulation of the primary market has become the preserve of SEBI. Further, the Ministry of Finance, Government of India, has transferred most of the powers under the Securities Contracts (Regulations) Act, 1956 to SEBI. The key SEBI guidelines are as follows:

- A company can make a 100 per cent retail issue provided it satisfies certain conditions relating to net tangible assets, track record of distributable profits, net worth, and issue size.

- In case an unlisted company does not satisfy any of the above conditions, it can make an IPO of equity shares or convertibles only if it meets certain conditions that relate to the use of the book building process, allotment to Qualified Institutional Buyers, participation by Financial Institution/Scheduled Banks, post-issue nominal value of equity capital, and compulsory market making.
- Every company, whether unlisted or listed, which is eligible to make a public issue, can freely price its shares.
- The mechanics for a public offer of a debt security are much the same as that of a public offer of equity. However, there are some differences.
- The price at which a preferential allotment of shares is made is subject to certain restrictions.

## SUMMARY

- The three broad areas of financial management are capital budgeting, capital structure, and working capital management.
- Finance theory, in general, rests on the premise that the goal of financial management should be to maximise the wealth of shareholders. This goal is conceptually superior to alternatives like maximisation of profit, maximisation of earnings per share, and maximisation of return on equity.
- A business proposal augments the value of the firm if its net present value is positive.
- In many ways, financial management is an integral part of the jobs of managers who are involved in planning, resource allocation, and control. There are, however, many tasks of financial management and allied areas (like accounting) which are specialised in nature and attended to by the two key financial officers of the firm, *viz.* the treasurer and the controller.
- A basic knowledge of macro-economics is necessary for understanding the environment in which the firm operates and a good grasp of micro-economic principles is helpful in sharpening the tools of financial decision making.
- Accounting is concerned with score keeping, whereas finance is aimed at value maximising; accounting relies on the accrual method, whereas finance is based on the cash flow method; accounting deals with certainties, whereas finance copes with uncertainties.
- Corporate investment and financing decisions are guided, shaped, and circumscribed by a regulatory framework. The principal elements of this regulatory framework are: (a) Industrial Policy, (b) The Companies Act, and (c) Securities and Exchange Board of India Guidelines.

## QUESTIONS

1. Discuss the three broad areas of financial decision making.
2. What is the justification for the goal of maximising the wealth of shareholders?
3. Critically evaluate the goals of maximisation of profit and maximisation of return on equity.
4. What is the fundamental principle of finance?

### **1.16** *Fundamentals of Financial Management*

5. Describe the key activities of financial management.
6. How is the finance function organised in a large company?
7. Discuss the relationship of financial management to economics and accounting.
8. Discuss the key elements of the regulatory framework applicable to corporate investment and financing decisions.

# Financial Planning and Forecasting

## LEARNING OBJECTIVES

After studying this chapter, you should be able to:

- Describe certain common elements of financial plans
- Explain the per cent of sales method and the budget expense method for preparing the *pro forma* profit and loss account
- Discuss the formula for calculating additional funds needed
- Explain sustainable growth rate.

In the liberalised business environment, characterised by expanding opportunities and swiftly changing market conditions, strategic planning has assumed greater significance not only for growth but for survival as well.

The strategic plan of a firm spells out its corporate purpose, corporate scope, corporate objectives, and corporate strategies. The *corporate purpose* defines the mission of the firm. Put differently, it states the *raison d'être* of the firm. For example, a pharmaceutical company has expressed its purpose as "To provide quality health care for the masses." The *corporate scope* delineates the lines of business in which the firm wishes to engage in and the geographic spread of its operations. For example, Nucor Corporation, a firm listed on the New York Stock Exchange, has defined its scope as follows: "We are a manufacturing company, producing primarily steel products." While the corporate purpose and scope reflect the broad philosophy and nature of the firm's activities, *corporate objectives* spell out the specific goals sought by the firm. Some goals are stated in quantitative terms such as achieving a market share of 30 per cent, a growth rate of 25 per cent, and a return on equity of 20 per cent. Other goals are stated in qualitative terms such as "attaining a measure of stability." *Corporate strategies* or policies are the instruments for achieving corporate objectives. A firm may, for example, follow the policy of vertical integration to achieve growth and stability.

As a key member of the top management group that formulates the strategic plan, the financial manager must (i) sensitise the strategic planning group to the financial implications of various choices, (ii) ensure that the chosen strategic plan is financially

feasible, (iii) translate the strategic plan that is finally adopted—usually through an iterative procedure—into a long-range financial plan (typically a five year plan), and (iv) coordinate the development of the budget (which covers the first year of the long-range financial plan in great detail).

This chapter discusses various facets of financial planning and budgeting.

### 7.1 WHAT AND WHY OF FINANCIAL PLANNING

A long-term financial plan represents a blueprint of what a firm proposes to do in the future. Typically it covers a period of three to ten years—most commonly it spans a period of five years. Naturally, planning over such an extended time horizon tends to be in fairly aggregative terms. While there is considerable variation in the scope, degree of formality, and level of sophistication in financial planning across firms, most corporate financial plans have certain common elements.

- *Economic Assumptions* The financial plan is based on certain assumptions about the economic environment (interest rate, inflation rate, growth rate, exchange rate, and so on).
- *Sales Forecast* The sales forecast is typically the starting point of the financial forecasting exercise. Most financial variables are related to the sales figure.
- *Asset Requirements* Firms need to invest in plant and equipment and working capital. The financial plan spells out the projected capital investments and working capital requirements over time.
- *Financing Plan* Suitable sources of financing have to be thought of for supporting the investment in capital expenditure and working capital. The financing plan delineates the proposed means of financing.
- *Pro forma Statements* The heart of a financial plan is the *pro forma* (forecast) statement of profit and loss and balance sheets.
- *Cash Budget* The cash budget shows the cash inflows and outflows expected in the budget period.

Companies spend considerable time and resources in financial planning. Hence, it is reasonable to ask: What are the benefits of financial planning? *Inter alia*, financial planning:

- Identifies advance actions to be taken in various areas.
- Seeks to develop a number of options in various areas that can be exercised under different conditions.
- Facilitates a systematic exploration of interaction between investment and financing decisions.
- Clarifies the links between present and future decisions.
- Forecasts what is likely to happen in future and hence helps in avoiding surprises.
- Ensures that the strategic plan of the firm is financially viable.
- Provides benchmarks against which future performance may be measured.

## 7.2 SALES FORECAST

The sales forecast is typically the starting point of the financial forecasting exercise. Most of the financial variables are projected in relation to the estimated level of sales. Hence, the accuracy of the financial forecast depends critically on the accuracy of the revenues from operations (i.e. sales) forecast.

Although the financial manager may participate in the process of developing the sales forecast, the primary responsibility for it typically vests with the marketing department or the planning group.

Sales forecasts may be prepared for varying planning horizons to serve different purposes. A sales forecast for a period of 3–5 years, or for even longer durations, may be developed mainly to aid investment planning. A sales forecast for a period of one year (and in some cases two years) is the primary basis for the financial forecasting exercise discussed in this chapter. Sales forecasts for shorter durations (six months, three months, one month) may be prepared for facilitating working capital planning and cash budgeting.

A wide range of sales forecasting techniques and methods are available. They may be divided into three broad categories:

- *Qualitative Techniques* These techniques rely essentially on the judgment of experts to translate qualitative information into quantitative estimates.
- *Time Series Projection Methods* These methods generate forecasts on the basis of an analysis of the past behaviour of time series.
- *Causal Models* These techniques seek to develop forecasts based on cause-effect relationships expressed in an explicit, quantitative manner.

Each technique has its advantages and limitations. Often, exclusive reliance on a single technique is somewhat dangerous. Practical wisdom suggests that at least two techniques, which seem to make sense in the specific circumstances of the firm, may be employed.

## 7.3 PRO FORMA STATEMENT OF PROFIT AND LOSS

There are two commonly used methods for preparing the *pro forma* statement of profit and loss: the per cent of sales method and the budgeted expense method.

### Per Cent of Sales Method

The per cent of sales method for preparing the *pro forma* statement of profit and loss is fairly simple. Basically this method assumes that the future relationship between various elements of costs to sales will be similar to their historical relationship. When using this method, a decision has to be taken about which historical cost ratios to be used: Should these ratios pertain to the previous year, or the average of two or more previous years?

Exhibit 7.1 illustrates the application of the per cent of sales method for preparing the *pro forma* statement of profit and loss of Spaceage Electronics for the year 20X2. In this

## 7.4 Fundamentals of Financial Management

table historical data are given for two previous years, 20X0 and 20X1. For projection purposes, the average of the ratios of these two previous years has been used. The forecast value of each item is obtained as the product of the estimated sales and the average per cent of sales ratio applicable to that item. For example, the average per cent of sales ratio for material expenses is 45.84 per cent. Multiplying the estimated sales of 1400 by 45.84 per cent, the projected value of material expenses has been calculated. Likewise, the projected values of other items in the income statement have been calculated. Although in practice some deviation from a mechanical application of this method is unavoidable, for the sake of illustration the projections shown in Exhibit 7.1 are based on a strict application of this method, except for dividends and retained earnings. Remember that the distribution of earnings between dividends and retained earnings reflects a managerial policy which is not easily expressible in mechanistic terms.

### Budgeted Expense Method

The per cent of sales method, though simple, is too rigid and mechanistic. For deriving the *pro forma* statement of profit and loss, shown in Exhibit 7.1, we assumed that all elements of costs and expenses bore a strictly proportional relationship to sales. The budgeted expense method, on the other hand, calls for estimating the value of each item on the basis of expected developments in the future period for which the *pro forma* statement of profit and loss is being prepared. Obviously, this method requires greater effort on the part of management because it calls for defining likely developments.

**Exhibit 7.1** Pro Forma Statement of Profit and for Spaceage Electronics for 20X2 Based on Per Cent of Sales Method

|                                                     | Historical data |      | Average<br>per cent of<br>sales | Pro forma statement of<br>profit and loss of 20X2<br>assuming revenues from<br>operations of 1400 |
|-----------------------------------------------------|-----------------|------|---------------------------------|---------------------------------------------------------------------------------------------------|
|                                                     | 20X0            | 20X1 |                                 |                                                                                                   |
| Revenues from Operations                            | 1200            | 1280 | 100                             | 1400                                                                                              |
| Other Income                                        | 8               | 10   | 0.72                            | 10                                                                                                |
| Total Revenues                                      | 1208            | 1290 | 100.72                          | 1410                                                                                              |
| Expenses                                            |                 |      |                                 |                                                                                                   |
| Material expenses                                   | 547             | 590  | 45.84                           | 642                                                                                               |
| Employee benefit expenses                           | 274             | 295  | 22.94                           | 321                                                                                               |
| Finance costs                                       | 60              | 65   | 5.04                            | 71                                                                                                |
| Depreciation and amortisation<br>expenses           | 75              | 80   | 6.25                            | 88                                                                                                |
| Other expenses                                      | 98              | 103  | 8.11                            | 114                                                                                               |
| Total expenses                                      | 1054            | 1133 | 88.17                           | 1234                                                                                              |
| Profit before Exceptional Items and<br>Other Income | 154             | 157  | 12.55                           | 176                                                                                               |
| Exceptional Items                                   | 30              | 32   | 2.5                             | 35                                                                                                |

(Contd.)

|                                           | <i>Historical data</i> |             | <i>Average</i>                     | <i>Pro forma statement of</i>                                                                |
|-------------------------------------------|------------------------|-------------|------------------------------------|----------------------------------------------------------------------------------------------|
|                                           | <i>20X0</i>            | <i>20X1</i> | <i>per cent of</i><br><i>sales</i> | <i>profit and loss of 20X2</i><br><i>assuming revenues from</i><br><i>operations of 1400</i> |
| Profit before Extraordinary Items and Tax | 184                    | 189         | 15.05                              | 211                                                                                          |
| Extraordinary Items                       |                        |             |                                    |                                                                                              |
| Profit Before Tax                         | 184                    | 189         | 15.05                              | 211                                                                                          |
| Tax Expense                               | 82                     | 90          | 6.93                               | 97                                                                                           |
| Profit (Loss) for the Period              | 102                    | 99          | 8.12                               | 114                                                                                          |
| Dividends                                 | 60                     | 63          |                                    |                                                                                              |
| Retained Earnings                         | 42                     | 36          |                                    |                                                                                              |

## A Combination Method

It appears that a combination of the two methods described above often works best. For certain items which have a fairly stable relationship with sales, the per cent of sales method is quite adequate. For other items, where future is likely to be very different from the past, the budgeted expense method, which calls for managerial assessment of expected future developments, is eminently suitable. A combination method of this kind is neither overly simplistic as the per cent of sales method nor unduly onerous as the budgeted expense method.

Exhibit 7.2 presents the 20X2 *pro forma* statement of profit and loss for Spaceage Electronics, constructed by using a combination of the per cent of sales and the budgeted expense methods. Other income, material expenses, employee benefit expenses, finance costs and exceptional items are assumed to change proportionally with sales, the proportions being the average of the two preceding years. All the remaining items have been budgeted on some specific basis.

**Exhibit 7.2** Pro Forma Statement of Profit and Loss for Spaceage Electronics for 20X2  
Using the Combination Method

|                           | <i>Historical data</i> |             | <i>Average</i>                     | <i>Pro forma statement of</i>                                                                |
|---------------------------|------------------------|-------------|------------------------------------|----------------------------------------------------------------------------------------------|
|                           | <i>20X0</i>            | <i>20X1</i> | <i>per cent of</i><br><i>sales</i> | <i>profit and loss of 20X2</i><br><i>assuming revenues from</i><br><i>operations of 1400</i> |
| Revenues from Operations  | 1200                   | 1280        | 100                                | 1400                                                                                         |
| Other Income              | 8                      | 10          | 0.72                               | 10                                                                                           |
| Total Revenues            | 1208                   | 1290        | @                                  | 1410                                                                                         |
| Expenses                  |                        |             |                                    |                                                                                              |
| Material expenses         | 547                    | 590         | 45.84                              | 642                                                                                          |
| Employee benefit expenses | 274                    | 295         | 22.94                              | 321                                                                                          |
| Finance costs             | 60                     | 65          | 5.04                               | 71                                                                                           |

(Contd.)

## 7.6 Fundamentals of Financial Management

|                                                     | Historical data |      | Average<br>per cent of<br>sales | Pro forma statement of<br>profit and loss of 20X2<br>assuming revenues from<br>operations of 1400 |
|-----------------------------------------------------|-----------------|------|---------------------------------|---------------------------------------------------------------------------------------------------|
|                                                     | 20X0            | 20X1 |                                 |                                                                                                   |
| Depreciation and amortisation<br>expenses           | 75              | 80   | Budgeted                        | 85                                                                                                |
| Other expenses                                      | 98              | 103  | Budgeted                        | 107                                                                                               |
| Total Expenses                                      | 1054            | 1133 | @                               | 1226                                                                                              |
| Profit Before Exceptional Items and<br>other Income | 154             | 157  | @                               | 184                                                                                               |
| Exceptional Items                                   | 30              | 32   | 2.5                             | 35                                                                                                |
| Profit before Extraordinary Items<br>and Tax        | 184             | 189  | @                               | 219                                                                                               |
| Extraordinary Items                                 |                 |      |                                 |                                                                                                   |
| Profit Before Tax                                   | 184             | 189  | @                               | 219                                                                                               |
| Tax Expense                                         | 82              | 90   | Budgeted                        | 90                                                                                                |
| Profit (Loss) for the Period                        | 102             | 99   | @                               | 129                                                                                               |
| Dividends                                           | 60              | 63   | Budgeted                        | 70                                                                                                |
| Retained earnings                                   | 42              | 36   | @                               | 59                                                                                                |

@ These items are obtained using accounting identities.

## 7.4 PRO FORMA BALANCE SHEET

The projections of various items on the assets side and liabilities side of the balance sheet may be derived as follows:

1. Employ the per cent of sales method to project the items on the assets side, excepting 'investments' and 'miscellaneous expenditures and losses.'
2. Estimate the expected values for 'non-current investments' and 'long-term loans' and advances using specific information applicable to them.
3. Use the per cent of sales method to derive the projected values of all other balance sheet items.
4. Obtain the projected value of reserves and surplus by adding the projected retained earnings (from the *pro forma* statement of profit and loss) to the reserves and surplus figure of the previous period.
5. Set the projected values for equity capital to be tentatively equal to their previous values.
6. Compare the total of the assets side with that of the liabilities side and determine the balancing item (If assets exceed liabilities, the balancing item represents the 'external funds required.' If liabilities exceed assets, the balancing item represents the 'surplus available funds').

### Illustration

To illustrate the preparation of the *pro forma* balance sheet, let us continue with the example of Spaceage Electronics. Exhibit 7.3 shows the balance sheets of Spaceage for 20X0 and 20X1.

### Circularity Problem

A circularity problem arises when the *pro forma* financial statements are prepared because the statement of profit and loss and the balance sheet are interrelated. The *pro forma* balance sheet cannot be prepared unless the *pro forma* statement of profit and loss, showing the amount of retained earnings to be carried to the balance sheet, is ready. At the same time, without the *pro forma* balance sheet, we cannot figure out the finance cost associated with the amount of external financing, an item required to prepare the *pro forma* statement of profit and loss.

## 7.5 COMPUTERISED FINANCIAL PLANNING SYSTEMS

Computerised financial planning systems are gaining in popularity. At the heart of a computerised financial planning system is a model which specifies the relationships relevant to the firm. The model defines relationships like the following:

|                           |                                                     |
|---------------------------|-----------------------------------------------------|
| Material expenses         | = 65 per cent of revenues from operations           |
| Employee benefit expenses | = ₹100,000 + 5 per cent of revenues from operations |
| Inventories               | = 15 per cent of revenues from operations           |
| •                         | •                                                   |
| •                         | •                                                   |
| •                         | •                                                   |
| •                         | •                                                   |

A computerised financial planning system helps in preparing *pro forma* financial statements, estimating the requirement of external funds, and calculating a variety of ratios. Such a system naturally offers a number of advantages: (i) Once the model has been developed, the tedium of manual computations is eliminated. (ii) The circularity problem mentioned earlier is easily tackled as the computer can quickly perform the required iterations. (iii) The model can be employed very conveniently to perform sensitivity analysis.

Thanks to the above advantages, the computerised financial planning system strengthens considerably the firm's planning ability. However, there is a potential disadvantage associated with it which may be overlooked. The ease with which computations can be performed and forecasts generated may result in misdirected efforts. A large quantity of low-quality predictions may be churned out creating confusion and bewilderment on the part of management. Quality may be sacrificed at the altar of quantity. To guard against this danger, greater thought should be given to the kinds of scenarios evaluated and the quality of analysis.

## 7.8 Fundamentals of Financial Management

### Exhibit 7.3 Pro Forma Balance Sheet of Spaceage Electronics for December 31, 20X2

|                                | <i>Historical data</i> |             | <i>Average per<br/>cent of sales</i> | <i>Pro forma balance sheet<br/>of 20X2 assuming<br/>revenues from operations<br/>of 1400</i> |
|--------------------------------|------------------------|-------------|--------------------------------------|----------------------------------------------------------------------------------------------|
|                                | <i>20X0</i>            | <i>20X1</i> |                                      |                                                                                              |
| Revenues from Operations       | 1200                   | 1280        | 100                                  | 1400                                                                                         |
| <b>EQUITY AND LIABILITIES</b>  |                        |             |                                      |                                                                                              |
| Shareholders' Funds            |                        |             |                                      |                                                                                              |
| Share capital (Par value ₹10)  | 300                    | 300         | No change                            | 300                                                                                          |
| Reserves and surplus           | 250                    | 286         | Pro forma<br>statement<br>of P&L     | 309                                                                                          |
| Non-current Liabilities        |                        |             |                                      |                                                                                              |
| Long-term borrowings           | 500                    | 505         | 40.56                                | 568                                                                                          |
| Deferred tax liabilities (net) | 45                     | 50          | 3.83                                 | 54                                                                                           |
| Long-term provisions           | 55                     | 50          | 4.24                                 | 59                                                                                           |
| Current Liabilities            |                        |             |                                      |                                                                                              |
| Short-term borrowings          | 200                    | 200         | 16.15                                | 226                                                                                          |
| Trade payables                 | 100                    | 112         | 8.54                                 | 120                                                                                          |
| Other current liabilities      | 20                     | 30          | 2.01                                 | 28                                                                                           |
| Short-term provisions          | 30                     | 17          | 1.91                                 | 27                                                                                           |
| External Funds Requirement     |                        |             |                                      | 19                                                                                           |
|                                |                        |             |                                      | 1710                                                                                         |
| <b>ASSETS</b>                  |                        |             |                                      |                                                                                              |
| Non-current Assets             |                        |             |                                      |                                                                                              |
| Fixed assets                   | 750                    | 775         | 61.52                                | 861                                                                                          |
| Non-current investments        | 40                     | 40          | No<br>change                         | 40                                                                                           |
| Long-term loans and advances   | 60                     | 60          | No<br>change                         | 60                                                                                           |
| Current Assets                 |                        |             |                                      |                                                                                              |
| Current investments            | 30                     | 33          | 2.54                                 | 36                                                                                           |
| Inventories                    | 375                    | 380         | 30.47                                | 427                                                                                          |
| Trade receivables              | 200                    | 212         | 16.61                                | 233                                                                                          |
| Cash and cash equivalents      | 25                     | 28          | 2.14                                 | 30                                                                                           |
| Short-term loans and advances  | 20                     | 22          | 1.69                                 | 24                                                                                           |
|                                |                        |             |                                      | 1710                                                                                         |

## 7.6 ADDITIONAL FUNDS NEEDED

Most firms forecast their additional funds needed by constructing *pro forma* financial statements as described above. However, if the financial ratios of the firm remain constant, the additional funds needed can be forecast using the following formula:

$$\begin{array}{ccccccc} \text{Additional} & & \text{Required} & & \text{Spontaneous} & & \text{Increase} \\ \text{funds needed} & = & \text{increase} & - & \text{increase} & - & \text{in retained} \\ & & \text{in assets} & & \text{in liabilities} & & \text{earnings} \end{array}$$

$$\text{AFN} = (A^*/S_0) \Delta S - (L^*/S_0) \Delta S - mS_1r$$

where:  $A^*$  = assets that must increase if sales (revenues from operation) are to increase

$S_0$  = sales during the previous year

$\Delta S$  = change in sales expected

$L^*$  = liabilities that increase spontaneously with sales

$r$  = retention ratio

$S_1$  = sales expected during the next year

$m$  = net profit margin ratio

**Example** The following holds for Pioneer Chemicals:  $A^*/S_0 = 0.9$ ,  $\Delta S = ₹60$  million,  $L^*/S_0 = 0.40$ ,  $m = 0.05$ ,  $S_1 = ₹460$  million, and  $r = 0.4$ . What is the additional funds needed for Pioneer Chemicals for the forthcoming year?

The additional funds needed for Pioneer Chemicals are:

$$\begin{aligned} \text{AFN} &= (0.90)(60) - (0.40)(60) - (0.05)(460)(0.4) \\ &= ₹20.8 \text{ million} \end{aligned}$$

## 7.7 SUSTAINABLE GROWTH RATE

Growth is often the central theme of corporate planning. Firms generally state corporate goals in terms of growth rates.

While firms are interested in growth, they may be reluctant to raise external equity capital for the following reasons: (i) There may be significant degree of underpricing when external equity is raised. (ii) The cost of issue tends to be unreasonably high. (iii) The dilution of control, consequent to the external issue, may not be acceptable to the existing controlling interest.

Given managerial disinclination to raise external equity capital, the following question is often raised in corporate financial planning: What rate of growth can be sustained with internally available equity?

The sustainable growth rate is simply:

$$\text{Return on equity} \times \text{Retention ratio}$$

Since the return on equity is a product of net profit margin, asset turnover, and equity multiplier, the sustainable growth rate may be expressed as follows:

$$\begin{array}{ccccccc} \text{Sustainable} & = & \text{Net profit} & \times & \text{Asset} & \times & \text{Equity} \times \text{Retention} \\ \text{growth rate} & & \text{margin} & & \text{turnover} & & \text{multiplier} & \text{ratio} \end{array}$$

**Example** The following information is available for Astra Infotech Limited:

Net profit margin : 20%

## 7.10 Fundamentals of Financial Management

|                   |        |
|-------------------|--------|
| Asset turnover    | : 1.2  |
| Equity multiplier | : 1.1  |
| Retention rate    | : 0.80 |

What is the sustainable growth rate for Astra Infotech Limited?

The sustainable growth rate is:

$$20\% \times 1.2 \times 1.1 \times 0.80 = 21.1\%$$

This means that Astra Infotech Limited can grow at a rate of 21.1 per cent without resorting to external equity issue.

### SUMMARY

- The strategic plan of a firm spells out its corporate purpose, corporate scope, corporate objectives, and corporate strategies. The corporate purpose defines the mission of the firm; the corporate scope delineates the lines of business in which the firm wishes to engage in and the geographic spread of its operations; corporate objectives spell out the specific goals sought by the firm; corporate strategies are the instruments for achieving corporate objectives.
- The financial manager must (i) sensitise the strategic planning group to the financial implications of various choices, (ii) ensure that the chosen strategic plan is financially feasible, (iii) translate the strategic plan into a long-range financial plan, and (iv) coordinate the development of the budget.
- The long-term financial plan represents a blueprint of what the firm proposes to do in the future. Most corporate financial plans have the following elements: economic assumptions, sales forecast, *pro forma* financial statements, asset requirements, financing plan, and cash budget.
- *Inter alia*, financial planning identifies advance actions, develops options, forecasts what is likely to happen, and provides benchmarks against which future performance may be measured.
- The sales forecast is typically the starting point of the financial forecasting exercise as most of the financial variables are projected in relation to the estimated level of sales.
- A wide range of sales forecasting methods are available. They may be divided into three broad categories: qualitative techniques, time series projection methods, and causal models.
- There are two commonly used methods for preparing the *pro forma* statement of profit and loss: the per cent of sales method and the budgeted expense method.
- The per cent of sales method assumes that the future relationship between various elements of costs to sales will be similar to their historical relationship. The budgeted expense method calls for estimating the value of each item on the basis of expected developments in the future period. It appears that a combination of the two methods often works best.
- The *pro forma* balance sheet relies mainly on the percent of sales method.
- If the financial ratios of the firm remain constant, the additional funds needed can be forecast using the formula.

|                         |   |                             |   |                                     |   |                               |
|-------------------------|---|-----------------------------|---|-------------------------------------|---|-------------------------------|
| Additional funds needed | = | Required increase in assets | – | Spontaneous increase in liabilities | – | Increase in retained earnings |
|-------------------------|---|-----------------------------|---|-------------------------------------|---|-------------------------------|

- Sustainable growth rate is equal to:

$$\text{Net profit margin} \times \text{Asset turnover} \times \text{Equity multiplier} \times \text{Retention ratio}$$

## QUESTIONS

1. Describe what is meant by corporate purpose, corporate scope, corporate objectives, and corporate strategies.
2. What are the elements of a corporate financial plan?
3. What are the benefits of financial planning?
4. Describe briefly the types of sales forecasting techniques and methods.
5. Explain the following methods for preparing the *pro forma* income statement: (i) the per cent of sales method, (ii) the budgeted expense method, and (iii) the combination method.
6. Describe the procedure for preparing the *pro forma* balance sheet.
7. Comment on the circularity problem on account of the income statement and the balance sheet being interrelated.
8. How would you estimate 'additional funds needed'?
9. What rate of growth can be sustained with internally available equity?
10. What is a computerised financial planning system or model? What are its advantages and disadvantages?

## SOLVED PROBLEMS

**SP7.1** The income statements and balance sheets of Deepam Silks for years 1 and 2 are as follows:

Income Statement

|                                                  | Year 1 | Year 2 |
|--------------------------------------------------|--------|--------|
| Revenues from Operations                         | 600    | 720    |
| Expenses                                         |        |        |
| Material expenses                                | 300    | 344    |
| Employee benefit expenses                        | 150    | 172    |
| Finance costs                                    | 10     | 12     |
| Depreciation and amortisation expenses           | 30     | 40     |
| Other expenses                                   | 86     | 100    |
| Total Expenses                                   | 576    | 668    |
| Profit before Exceptional Items and Other Income | 24     | 52     |
| Exceptional Items                                | 10     | 8      |
| Profit before Extraordinary Items and Tax        | 34     | 60     |
| Extraordinary Items                              |        |        |
| Profit Before Tax                                | 34     | 60     |
| Tax Expense                                      | 14     | 26     |
| Profit (Loss) for the period                     | 20     | 34     |
| Dividends                                        | 12     | 15     |
| Retained earnings                                | 8      | 19     |

## 7.12 Fundamentals of Financial Management

Balance Sheet

|                                | Year 1     | Year 2     |
|--------------------------------|------------|------------|
| <b>EQUITY AND LIABILITIES</b>  |            |            |
| <b>Shareholders' Funds</b>     |            |            |
| Share capital (Par value ₹10)  | 120        | 120        |
| Reserves and surplus           | 150        | 169        |
| <b>Non-current Liabilities</b> |            |            |
| Long-term borrowings           | 35         | 58         |
| Deferred tax liabilities (net) |            |            |
| Long-term provisions           | 5          | 11         |
| <b>Current Liabilities</b>     |            |            |
| Short-term borrowings          | 20         | 22         |
| Trade payables                 | 125        | 130        |
| Other current liabilities      | 5          | 5          |
| Short-term provisions          | 40         | 45         |
|                                | <u>500</u> | <u>560</u> |
| <b>ASSETS</b>                  |            |            |
| <b>Non-current Assets</b>      |            |            |
| Fixed assets                   | 240        | 270        |
| Non-current investments        | 7          | 8          |
| Long-term loans and advances   | 20         | 18         |
| <b>Current Assets</b>          |            |            |
| Current investments            | 3          | 2          |
| Inventories                    | 125        | 144        |
| Trade receivables              | 80         | 90         |
| Cash and cash equivalents      | 5          | 6          |
| Short-term loans and advances  | 20         | 22         |
|                                | <u>500</u> | <u>560</u> |

Prepare the *pro forma* income statement for year 3 and the *pro forma* balance sheet as at the end of year 3, based on the following assumptions:

- The projected sales for year 3 are 850
- The forecast values for the following income statement items may be derived using the per cent of sales method (for this purpose, assume that the average of the percentages for years 1 and 2 is applicable).
  - Material expenses
  - Employee benefit expenses
  - Finance costs
  - Other expenses
  - Exceptional items
- The forecast values for the other items of the income statement are as follows.
  - Depreciation : 45

- Tax : 50 per cent of earnings before tax
- Dividends : 16

(d) The forecast values of various balance sheet items may be derived as follows:

- Fixed assets (net) : Budgeted at 300
- Non-current investments : No change over year 2
- Current assets : Per cent of sales method wherein the percentages are based on the average for the previous two years.
- Share capital : No change over year 2
- Reserves and surplus : *Pro forma* income statement
- Non-current and current liabilities and provisions : Per cent of sales method wherein the percentages are based on the average for the previous two years.
- External fund required : Balancing item

#### Solution

The *pro forma* income statement and the *pro forma* balance sheet are shown below:

Pro forma Income Statement Deepam Silks for Year 3

|                                                  | Historical data |        | Average           | Pro forma income     |
|--------------------------------------------------|-----------------|--------|-------------------|----------------------|
|                                                  | Year 1          | Year 2 | Per cent of sales | statement for year 3 |
| Revenues from Operations                         | 600             | 720    | 100%              | 850                  |
| Expenses                                         |                 |        |                   |                      |
| Material expenses                                | 300             | 344    | 48.89             | 416                  |
| Employee benefit expenses                        | 150             | 172    | 24.44             | 208                  |
| Finance costs                                    | 10              | 12     | 1.67              | 14                   |
| Depreciation and amortisation expenses           | 30              | 40     | budgeted          | 45                   |
| Other expenses                                   | 86              | 100    | 14.11             | 120                  |
| Total Expenses                                   | 576             | 668    | @                 | 803                  |
| Profit before Exceptional items and Other Income | 24              | 52     | @                 | 47                   |
| Exceptional Items                                | 10              | 8      | 1.39              | 12                   |
| Profit before Extraordinary Items and Tax        | 34              | 60     | @                 | 59                   |
| Extraordinary Items                              |                 |        |                   |                      |
| Profit Before Tax                                | 34              | 60     | @                 | 59                   |
| Tax Expense                                      | 14              | 26     | @50% of PBT       | 30                   |
| Profit (Loss) for the Period                     | 20              | 34     | @                 | 29                   |
| Dividends                                        | 12              | 15     | budgeted          | 16                   |
| Retained Earnings                                | 8               | 19     | @                 | 13                   |

## 7.14 Fundamentals of Financial Management

Pro forma Balance Sheet for Deepam Silks for Year 3

|                                | <i>Historical data</i> |               | <i>Average per cent of sales or some other basis</i> | <i>Projection for year 3 based on a forecast sales of 850</i> |
|--------------------------------|------------------------|---------------|------------------------------------------------------|---------------------------------------------------------------|
|                                | <i>Year 1</i>          | <i>Year 2</i> |                                                      |                                                               |
| Revenues from Operations       | 600                    | 720           | 100                                                  | 850                                                           |
| EQUITY AND LIABILITIES         |                        |               |                                                      |                                                               |
| Shareholders' Funds            | 634                    |               |                                                      |                                                               |
| Share capital (Par value ₹10)  | 120                    | 120           | No change                                            | 120                                                           |
| Reserves and surplus           | 150                    | 169           | Pro forma statement of P&L                           | 182                                                           |
| Non-current Liabilities        |                        |               |                                                      |                                                               |
| Long-term borrowings           | 35                     | 58            | 6.94                                                 | 59                                                            |
| Deferred tax liabilities (net) |                        |               |                                                      | 0                                                             |
| Long-term provisions           | 5                      | 11            | 1.18                                                 | 10                                                            |
| Current Liabilities            |                        |               |                                                      |                                                               |
| Short-term borrowings          | 20                     | 22            | 3.19                                                 | 27                                                            |
| Trade payables                 | 125                    | 130           | 19.44                                                | 165                                                           |
| Other current liabilities      | 5                      | 5             | 0.76                                                 | 6                                                             |
| Short-term provisions          | 40                     | 45            | 6.46                                                 | 55                                                            |
| External funds requirement     |                        |               |                                                      | 23                                                            |
| ASSETS                         | 500                    | 560           |                                                      | 647                                                           |
| Non-current Assets             |                        |               |                                                      |                                                               |
| Fixed assets                   | 240                    | 270           | budgeted                                             | 300                                                           |
| Non-current investments        | 7                      | 8             | no change                                            | 8                                                             |
| Long-term loans and advances   | 20                     | 18            | no change                                            | 18                                                            |
| Current Assets                 |                        |               |                                                      |                                                               |
| Current investments            | 3                      | 2             | 0.39                                                 | 3                                                             |
| Inventories                    | 125                    | 144           | 20.42                                                | 174                                                           |
| Trade receivables              | 80                     | 90            | 12.92                                                | 110                                                           |
| Cash and cash equivalents      | 5                      | 6             | 0.83                                                 | 7                                                             |
| Short-term loans and advances  | 20                     | 22            | 3.19                                                 | 27                                                            |
|                                | 500                    | 560           |                                                      | 647                                                           |

## PROBLEMS

1. The income statement of Modern Electronics Limited for years 1 and 2 is as follows:

|                                                  | Year |     |
|--------------------------------------------------|------|-----|
|                                                  | 1    | 2   |
| Revenues from Operations                         | 800  | 890 |
| Expenses                                         |      |     |
| Material expenses                                | 407  | 453 |
| Employee benefit expenses                        | 203  | 227 |
| Finance costs                                    | 10   | 11  |
| Depreciation and amortisation expenses           | 50   | 64  |
| Other expenses                                   | 120  | 117 |
| Total Expenses                                   | 790  | 872 |
| Profit before Exceptional Items and Other Income | 10   | 18  |
| Exceptional Items                                | 8    | 10  |
| Profit before Extraordinary Items and Tax        | 18   | 28  |
| Extraordinary Items                              |      |     |
| Profit before Tax                                | 18   | 28  |
| Tax Expense                                      | 7    | 10  |
| Profit (Loss) for the period                     | 11   | 18  |
| Dividends                                        | 6    | 7   |
| Retained earnings                                | 5    | 11  |

Using the per cent of sales method, prepare the *pro forma* income statement for year 3. Assume that the sales will be 1020 in year 3. If dividends are raised to 8, what amount of retained earnings can be expected for year 3?

2. Rework Problem 1 assuming the following budgeted amounts.

|                                        |     |
|----------------------------------------|-----|
| General and administration expenses    | 124 |
| Depreciation and amortisation expenses | 60  |
| Finance costs                          | 12  |
| Dividends                              | 8   |

3. The balance sheet of Modern Electronics Ltd. as of the end of years 1 and 2 is given below:

|                               | Year |     |
|-------------------------------|------|-----|
|                               | 1    | 2   |
| EQUITY AND LIABILITIES        |      |     |
| Shareholders' Funds           |      |     |
| Share capital (Par value ₹10) | 150  | 150 |
| Reserves and surplus          | 118  | 129 |
| Non-current Liabilities       |      |     |
| Long-term borrowings          | 144  | 175 |

(Contd.)

## 7.16 Fundamentals of Financial Management

|                               | Year |     |
|-------------------------------|------|-----|
|                               | 1    | 2   |
| Long-term provisions          | 13   | 19  |
| Current Liabilities           |      |     |
| Short-term borrowings         | 150  | 180 |
| Trade payables                | 126  | 167 |
| Short-term provisions         | 40   | 45  |
|                               | 741  | 865 |
| ASSETS                        |      |     |
| Non-current Assets            |      |     |
| Fixed assets                  | 300  | 380 |
| Non-current investments       | 20   | 20  |
| Long-term loans and advances  | 15   | 14  |
| Current Assets                |      |     |
| Current investments           | 21   | 20  |
| Inventories                   | 173  | 192 |
| Trade receivables             | 180  | 200 |
| Cash and cash equivalents     | 12   | 14  |
| Short-term loans and advances | 20   | 25  |
|                               | 741  | 865 |

As in Problem 1, assume that sales will grow to 1020 in year 3. For the year 3, on the asset side assume that current and non-current investments would remain unchanged and on the equities and liabilities assume that the share capital and long-term provisions would remain unchanged. Other items in the balance sheet would change in proportion to sales.

Obtain the estimated value of retained earnings from the *pro forma* income statement developed in Problem 2.

Finally, estimate the amount of external financing needed for year 3.

**LEARNING OBJECTIVES**

After studying this chapter, you should be able to:

- Describe the possible motives for holding cash
- Explain the receipts and payments method for cash budgeting
- Suggest methods to enhance the efficiency of cash management
- Discuss the options available for deploying short-term surplus funds

Cash, the most liquid asset, is of vital importance to the daily operations of business firms. While the proportion of corporate assets held in the form of cash is very small, often between 1 per cent and 3 per cent, its efficient management is crucial to the solvency of the business because in a very important sense cash is the focal point of fund flows in a business. In view of its importance, it is generally referred to as the “life blood of a business enterprise”.

Why does a firm need cash? As John Maynard Keynes put forth, there are three possible motives for holding cash.

**Transaction Motive** Firms need cash to meet their transaction needs. The collection of cash (from sale of goods and services, sale of assets, and additional financing) is not perfectly synchronised with the disbursement of cash (for purchase of goods and services, acquisition of capital assets, and meeting other obligations). Hence, some cash balance is required as a buffer.

**Precautionary Motive** There may be some uncertainty about the magnitude and timing of cash inflows from sale of goods and services, sale of assets, and issuance of securities. Likewise, there may be uncertainty about cash outflows on account of purchases and other obligations. To protect itself against such uncertainties, a firm may require some cash balance.

**Speculative Motive** Firms would like to tap profit making opportunities arising from fluctuations in commodity prices, security prices, interest rates, and foreign exchange rates. A cash-rich firm is better prepared to exploit such bargains. Hence, firms which

### 23.2 Fundamentals of Financial Management

have such speculative leanings may carry additional liquidity. However, for most firms their reserve borrowing capacity and marketable securities would suffice to meet their speculative needs.

While cash serves these functions, it is an idle resource which has an opportunity cost. The liquidity provided by cash holding is at the expense of profits sacrificed by foregoing alternative investment opportunities. Hence, the financial manager should (i) establish reliable forecasting and reporting systems, (ii) improve cash collections and disbursements, and (iii) achieve optimal conservation and utilisation of funds.

## 23.1 CASH BUDGETING

Cash budgeting or short-term cash forecasting is the principal tool of cash management. Cash budgets, routinely prepared by business firms, are helpful in: (i) estimating cash requirements, (ii) planning short-term financing, (iii) scheduling payments in connection with capital expenditure projects, (iv) planning purchases of materials, (v) developing credit policies, and (vi) checking the accuracy of long-term forecasts.

Firms use multiple short-term forecasts of varying length and detail suited to meet different needs. The commonly used designs for short-term cash forecasts are: (i) one year divided into quarters or months, (ii) one quarter divided into months, and (iii) one month divided into weeks. A firm, hard pressed with liquidity crunch, may even prepare a weekly cash forecast divided into days.<sup>1</sup> The point to be emphasised here is that these multiple formats serve differing purposes and should not be regarded as mutually exclusive.

The principal method of short-term cash forecasting is the receipts and payments method. Sometimes the adjusted net income method is used though this method is employed mainly for long-term cash forecasting.

### Receipts and Payments Method

The cash budget prepared under this method shows the timing and magnitude of expected cash receipts and payments over the forecast period. It includes all expected receipts and payments irrespective of how they are classified in accounting. The items of cash receipts and cash payments and the bases for estimating them are shown in Exhibit 23.1.

From Exhibit 23.1 it is clear that the receipts and payments method of cash forecasting calls for information about estimated sales, production plan, purchasing plan, financing plan, and capital expenditure budget. The most crucial input in the entire process, of course, is the figure of estimated sales because various business plans are closely related to estimated sales.

**Illustration** The preparation of cash budget may be illustrated with an example. ABC Company manufactures razor blades. Its estimated sales for the period January 20X0 through June 20X0 are: ₹100,000 per month from January through March and ₹120,000 per month from April through June. The sales for November and December of the previous

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<sup>1</sup> Of course, I would not like to press the point further for an already hard pressed firm!

**Exhibit 23.1** Items of Cash Receipts and Payments and the Basis of Their Estimation

| <i>Items</i>                                       | <i>Basis of Estimation</i>                                                                        |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Cash sales                                         | Estimated sales and its division between cash and credit sales                                    |
| Collection of accounts receivable                  | Estimated sales, its division between cash and credit sales, and collection pattern               |
| Interest and dividend receipts                     | Firm's portfolio of securities and return expected from the portfolio                             |
| Increase in loans/deposits and issue of securities | Financing plan                                                                                    |
| Sale of assets                                     | Proposed disposal of assets                                                                       |
| Cash purchases                                     | Estimated purchases, and its division between cash and credit purchases.                          |
| Payment for purchases                              | Estimated purchases, its division between cash purchases and credit purchases, and payment terms. |
| Wages and salaries                                 | Manpower employed and wages and salaries structure                                                |
| Manufacturing expenses                             | Production plan                                                                                   |
| General, administration and selling expenses       | Administration and sales personnel and proposed sales promotion and distribution expenditure      |
| Capital equipment purchases                        | Capital expenditure budget and payment pattern associated with capital equipment purchases        |
| Repayment of loans and retirement of securities    | Financing plan                                                                                    |

year have been ₹100,000 each. Cash and credit sales are expected to be 20 per cent and 80 per cent respectively. The receivables from credit sales are expected to be collected as follows: 50 per cent of receivables, on an average, one month from the date of sale and the balance 50 per cent, on an average, two months from the date of sale. No bad debt losses are expected to occur. Other anticipated receipts are: (i) ₹5,000 from the sale of a machine in March, and (ii) ₹2,000 interest on securities in June. Given this information, the forecasted cash receipts have been tabulated in Exhibit 23.2.

**Exhibit 23.2** Forecast of Cash Receipts

|                                      | <i>January</i> | <i>February</i> | <i>March</i> | <i>April</i> | <i>May</i> | (₹)<br><i>June</i> |
|--------------------------------------|----------------|-----------------|--------------|--------------|------------|--------------------|
| 1. Sales                             | 100,000        | 100,000         | 100,000      | 120,000      | 120,000    | 120,000            |
| 2. Credit Sales                      | 80,000         | 80,000          | 80,000       | 96,000       | 96,000     | 96,000             |
| 3. Collection of Accounts Receivable | 80,000         | 80,000          | 80,000       | 80,000       | 88,000     | 96,000             |
| 4. Cash Sales                        | 20,000         | 20,000          | 20,000       | 24,000       | 24,000     | 24,000             |
| 5. Receipt from Sale of Equipment    |                |                 | 5,000        |              |            |                    |
| 6. Interest                          |                |                 |              |              |            | 2,000              |
| Total Cash Receipts (3 + 4 + 5 + 6)  | 100,000        | 100,000         | 105,000      | 104,000      | 112,000    | 122,000            |

### 23.4 Fundamentals of Financial Management

We now consider the forecast of cash payments. ABC Company plans to purchase materials worth ₹40,000 in January and February and materials worth ₹48,000 each month from March through June. The payments for these purchases are made approximately a month after the purchase. The purchases for December of the previous year have been ₹40,000 for which payment will be made in January 20X0. Miscellaneous cash purchases of ₹2,000 per month are planned from January through June. Wage payments and manufacturing expenses are expected to be ₹15,000 and ₹20,000 per month respectively, January through June; general administrative and selling expenses are expected to be ₹10,000 per month. Dividend payment of ₹20,000 and tax payment of ₹20,000 are scheduled in June 20X0. A machine worth ₹50,000 is proposed to be purchased on cash in March 20X0. Given this information, the proposed payments are shown in Exhibit 23.3.

**Exhibit 23.3** Forecast of Cash Payments

|                                                   | January | February | March   | April  | May    | (₹)<br>June |
|---------------------------------------------------|---------|----------|---------|--------|--------|-------------|
| 1. Material Purchases                             | 40,000  | 40,000   | 48,000  | 48,000 | 48,000 | 48,000      |
| 2. Credit Material Purchases                      | 40,000  | 40,000   | 48,000  | 48,000 | 48,000 | 48,000      |
| 3. Payment of Accounts Payable                    | 40,000  | 40,000   | 40,000  | 48,000 | 48,000 | 48,000      |
| 4. Miscellaneous Cash Purchases                   | 2,000   | 2,000    | 2,000   | 2,000  | 2,000  | 2,000       |
| 5. Wages                                          | 15,000  | 15,000   | 15,000  | 15,000 | 15,000 | 15,000      |
| 6. Manufacturing Expenses                         | 20,000  | 20,000   | 20,000  | 20,000 | 20,000 | 20,000      |
| 7. General Administrative and<br>Selling Expenses | 10,000  | 10,000   | 10,000  | 10,000 | 10,000 | 10,000      |
| 8. Dividend                                       | —       | —        | —       | —      | —      | 20,000      |
| 9. Tax                                            | —       | —        | —       | —      | —      | 20,000      |
| 10. Capital Expenditure                           | —       | —        | 50,000  | —      | —      | —           |
| Total Payments                                    | 87,000  | 87,000   | 137,000 | 95,000 | 95,000 | 135,000     |
| (3 + 4 + 5 + 6 + 7 + 8 + 9 + 10)                  |         |          |         |        |        |             |

Assuming that the cash balance on 1st January 20X0 is ₹22,000 and the minimum cash balance required by the firm is ₹20,000, we can now prepare a summary statement. This statement, shown in Exhibit 23.4, calculates the surplus/deficit in relation to the minimum required. From Exhibit 23.4 it can be seen that a cash shortage is expected to occur during March and it would disappear in April as the business operations result in cash inflows. The shortage expected in March is due to the proposed capital expenditure of ₹50,000. The management can avoid this shortage by adopting one or more of the following means:<sup>2</sup> (i) postponement of asset acquisition to April, (ii) deferring a portion of the payment for the capital asset to April, and (iii) resorting to short-term borrowing for the month of March.

<sup>2</sup>Other alternatives like (i) delaying payment to suppliers of material and (ii) expediting the collection of receivables are also available.

**Exhibit 23.4** Summary Cash Forecast

|                                                                                    | January | February | March    | April   | May     | (₹)<br>June |
|------------------------------------------------------------------------------------|---------|----------|----------|---------|---------|-------------|
| 1. Opening Cash Balance                                                            | 22,000  |          |          |         |         |             |
| 2. Receipts                                                                        | 100,000 | 100,000  | 105,000  | 104,000 | 112,000 | 122,000     |
| 3. Payments                                                                        | 87,000  | 87,000   | 137,000  | 95,000  | 95,000  | 135,000     |
| 4. Net Cash Flow (2–3)                                                             | 13,000  | 13,000   | (32,000) | 9,000   | 17,000  | (13,000)    |
| 5. Cumulative Net Cash Flow                                                        | 13,000  | 26,000   | (6,000)  | 3,000   | 20,000  | 7,000       |
| 6. Opening Cash<br>Balance + Cumulative<br>Net Cash Flow (1 + 5)                   | 35,000  | 48,000   | 16,000   | 25,000  | 42,000  | 29,000      |
| 7. Minimum Cash Balance Required                                                   | 20,000  | 20,000   | 20,000   | 20,000  | 20,000  | 20,000      |
| 8. Surplus or Deficit in Relation to the<br>Minimum Cash Balance<br>Required (6–7) | 15,000  | 28,000   | (4,000)  | 5,000   | 22,000  | 9,000       |

**Deviations from Expected Cash Flows**

Expressed as it is in numbers, the cash budget often conveys a picture of precision. Hence, a great deal of faith is usually put on it. A moment's reflection, however, would reveal that the figures found in the cash budget merely represent *estimates* of future cash flows. The actual cash flows are likely to deviate from the estimates provided in the cash budget—the extent of deviation depends mainly on how volatile the cash flows of the business are.

Given the uncertainties characterising business operations, estimating the cash flows on the basis of a single set of assumptions, as is done in conventional cash budgeting, often results in an inadequate perspective on the future. Hence, it is advisable to prepare additional cash budgets, based on different sets of assumptions. The least that a firm can do is to look at cash forecasts under three possible scenarios: pessimistic scenario, normal scenario, and optimistic scenario. Such an analysis provides a better perspective on future cash flows and facilitates the formulation of contingency plans.

**Evaluation**

The receipts and payments method of cash forecasting is used commonly because of two advantages: (i) It provides a complete picture of expected cash flows. (ii) It is a sound vehicle for exercising control over day-to-day transactions. This method, however, has some drawbacks too: (i) Its reliability is impaired by delays in collection or sudden demand for large payments and other similar factors. (ii) It fails to provide a clear picture of important changes in the company's working capital movement, especially those relating to inventories and receivables.

**23.2 LONG-TERM CASH FORECASTING**

The dividing line between short-term cash forecasts and long-term cash forecasts is usually one year, though the distinction is somewhat arbitrary. Long-term cash forecasts are generally prepared for a period ranging from two to five years and serve to provide a

### 23.6 Fundamentals of Financial Management

broad brush picture of a firm's financing needs and availability of investible surplus in future. Such forecasts are helpful in planning capital investment outlays and long-term financing.

While the receipts and disbursements methods can theoretically be used for preparing the long-term cash forecast, the method that is generally used for this purpose is the *adjusted net income method*. This method of cash forecasting, resembling the funds flow statement, seeks to estimate the firm's need for cash at some future date and indicate whether this need can be met from internal sources or not. A format for preparing the adjusted net income forecast is shown in Exhibit 23.5. The information required for preparing the adjusted net income forecast is derived mainly from the budgets prepared by the firm.

**Exhibit 23.5** Format for the Adjusted Net Income Method

|                                    | 20X0 | 20X1 | 20X2 | 20X3 | 20X4 |
|------------------------------------|------|------|------|------|------|
| <i>Sources</i>                     |      |      |      |      |      |
| Net Income after Taxes             |      |      |      |      |      |
| Non-cash Charges                   |      |      |      |      |      |
| (Depreciation, Amortisation, etc.) |      |      |      |      |      |
| Increase in Borrowings             |      |      |      |      |      |
| Sale of Equity Shares              |      |      |      |      |      |
| Miscellaneous                      |      |      |      |      |      |
| <i>Uses</i>                        |      |      |      |      |      |
| Capital Expenditures               |      |      |      |      |      |
| Increase in Current Assets         |      |      |      |      |      |
| Repayment of Borrowings            |      |      |      |      |      |
| Dividend Payment                   |      |      |      |      |      |
| Miscellaneous                      |      |      |      |      |      |
| <i>Surplus/Deficit</i>             |      |      |      |      |      |
| Opening Cash Balance               |      |      |      |      |      |
| Closing Cash Balance               |      |      |      |      |      |

## 23.3 MONITORING COLLECTIONS AND RECEIVABLES

To enhance the efficiency of cash management, collections and disbursements must be properly monitored. In this respect, the following are helpful.

### Prompt Billing

Often there is a time lag between the despatch of goods or provision of service and the sending of bills. By preparing and sending the bills promptly, a firm can ensure earlier remittance. It should be realised that it is in the area of billing that the company's control is high and there is a sizeable opportunity to free up cash. To tap this opportunity the treasurer should work with the controller and others in (i) accelerating invoice data, (ii) mailing bills promptly, and (iii) identifying payment locations.